Functional Requirement Specification
for
ANNA DARPAN
(Ticket Mechanism Module)

Development Object ID	**:**	**FRS_AnnaDarpan_Ticket_Mechanism_V0.7**
Development Object Title	**:**	**Ticket Mechanism**
Version	**:**	**0.7**
Document Status	**:**	**Draft Version**

Table of Contents

Sign-off & Revision History

Document
Internal/Peer Review Stakeholder Review

Functional Requirement Specification (FRS)	
Status:	**In Progress,** Completed**,** signed off, Cancelled
Version	0.7

FS Sign-Off			
Responsibility	**Name**	**Signature**	**Date**

Record of Release		
Version No	**Date of Change**	**Description of Changes**
0.1	16-10-2024	Initial FRS document for Anna Darpan on Ticket Mechanism module.
0.2	13-11-2024	Incorporated the changes suggested by Stakeholder on 5/11/2024 against the FRS submitted
0.3	12-12-2024	Incorporated the changes as per the feedback received on 28-11-2024
0.4	17-01-2025	Updated SLA and Root cause analysis as per RFP
0.5	20-02-2025	Updated as per the feedback received on 07-02-2025
0.6	02-05-2025	Updated as per the feedback received by PMU on 2-05-2025
0.7	05-05-2025	Incorporated the feedback received from PMU and CGM Sir on 5th May 2025

Priority	Low / Medium / **High**

Reason:

i. **L1 (Level 1)**: Basic support, typically the first point of contact. Handles simple and frequently encountered issues, like initial troubleshooting, and basic guidance.
ii. **L2 (Level 2)**: More specialized support, addressing more complex technical issues that require deeper investigation.
iii. **L3 (Level 3)**: The most advanced support, involving experts or development teams.

Complexity	Low / **Medium** / High

Reason:

i. Ticket Creation and Assignment
ii. Ticket Resolution and tracking

Abbreviations & Acronyms

S. No.	Abbreviation	Full Form
1	API	Application Programming Interface
2	BA	Business Analyst
3	DBA	Database Administrator
4	DO	Divisional Office
5	FCI	Food Corporation of India
6	L1, L2, L3	Level 1, Level 2, Level 3
7	MIS	Management Information System
8	P1, P2, P3	Priority 1, Priority 2, Priority 3
9	RFP	Request For Proposal
10	RO	Regional Office
11	SLA	Service Level Agreement
12	SRS	System Requirement Specification
13	SOP	Standard Operating Procedure
14	ZO	Zonal Office

1. Introduction

A comprehensive solution is designed to handle service requests and incidents raised by end-users and manage their resolution through different support levels (L1, L2, and L3). The system aims to automate and streamline the workflow of creating, classifying, escalating, resolving, and closing tickets, providing a structured approach to incident management and enhancing operational efficiency.

This Functional Requirements Specification (FRS) outlines the key functionalities, user roles, and system interactions necessary for the effective implementation of the Ticket Mechanism at FCI. It aims to provide a comprehensive understanding of the requirements needed to develop a solution that aligns with FCI's operational goals and user expectations.

1.1 Objective:

The primary objective of **Ticket Mechanism module** is to provide an efficient, organized, and scalable solution for managing and automating the process of handling customer support requests. By centralizing communication between customers and support teams, it aims to streamline the lifecycle of service requests (tickets), ensuring quick resolution, maintaining service quality, and enhancing customer satisfaction.

1.2 Purpose of the Document

This document defines the functional requirements of Ticket Mechanism for managing tickets, incidents, and requests across multiple support levels. It ensures all business requirements are captured and defines how these are translated into technical functionalities.

Key objectives of this document include:

i. **Foundation for System Requirement Specification (SRS)**:
This document would serve as the basis for preparing the System Requirement Specification (SRS) document, ensuring that the development aligns with business needs and functional objectives.

ii. **Reference for Standard Operating Procedures (SOPs)**:
It would also be a key reference for structuring the project's Standard Operating Procedures (SOPs), providing a clear framework for operational workflows, digitalization, and process modernization.

iii. **Audience**:
The intended audience of this document includes all project stakeholders from FCI, as well as the team members from Coforge Ltd. including Program Manager, Solution Architect, Domain Expert, Technical Lead, Functional Tester, IT Infrastructure Lead, Senior Business Analyst, Data Architect, UI Lead, BA, App DBA, Developers, who would be responsible for implementing the functional requirements into the system.

Nomenclature Used in the Document:

In the context of this document, the following terminology has been used:

i. **Food Corporation of India (FCI)**:
The term "FCI" is used to collectively refer to the Food Corporation of India and all key stakeholders involved in the project from FCI's perspective.

ii. **Coforge Ltd. (Coforge)**:
Coforge Ltd., the technical solution provider for this project, is referred to as "Coforge" throughout the document.

This purpose and nomenclature ensure clarity and consistency across the document, providing a solid foundation for both project development and operational execution.

1.3 Scope

The scope of this system includes:

i. Self-ticket creation is enabled for Anna Darpan users. Tickets can also be created by a helpdesk representative on behalf of End user, after receiving a call/email from an Anna Darpan end User.
ii. The system should have the capability to record calls made to the helpdesk. The recorded calls should be retained for a specified period before being automatically refreshed or deleted to manage storage efficiently.
iii. Ticket Ids can be tracked efficiently.
iv. Automated workflows are established between L1, L2, and L3 support teams.
v. SLA (Service Level Agreement) management is implemented to ensure compliance.
vi. Availability of knowledge base, for L1, L2, L3 support team, who can access the relevant documents (including SOPs, guidelines, User Manuals, FAQ)
vii. An escalation mechanism should be in place in case of SLA breach.
viii. The system should facilitate comprehensive ticket analysis to identify and resolve underlying issues. This involves conducting Root Cause Analysis (RCA) for significant tickets to ensure that recurring problems are addressed effectively.
ix. Reporting are available in real time to monitor performance and ticket resolution times.
x. Alerts or notifications should be implemented for Anna Darpan End users to update the ticket status.
xi. System should also capture the satisfaction level/feedback of the End-user on ticket resolution.

1.4 Stakeholder

Though all Anna Darpan end-users are stakeholders in the ticket mechanism module, a few key stakeholders are mentioned below:

i. **FCI Headquarters (HQ):**
   a. HQ use to monitor ticket volumes, performance metrics, and SLA compliance across all offices and depots.

b. HQ might create tickets for escalated issues that need attention from Zonal or Regional Offices, especially for system-wide concerns.

ii. **Zonal Office (ZO):**
   a. Zonal Offices track the resolution of tickets within their jurisdiction.
   b. Zonal Offices create tickets for system issues.

iii. **Regional Office (RO):**
   a. RO use to track response and resolution times.
   b. Regional Offices create tickets for support requests originating from customers, depot issues, or internal operational concerns.

iv. **Divisional Office (DO):**
   a. DO monitors the Ticket raised and resolution times, within their jurisdiction.
   b. Divisional Offices create tickets for system related issues.

v. **Depot:**
   a. Create tickets for operational issues.
   b. Provision to track the tickets and re-open the ticket (if required).

vi. **Miller:**
   a. Millers create tickets for login or application-related issues.
   b. Provision to track the tickets and re-open the ticket (if required).

vii. **Contractors, Farmers, Agencies, etc.:**
   a. Provision to create, track and re-open ticket.

Disclaimer: Roles and permissions are managed through the User Management module, so the specific rights as per role are not detailed above.

1.5 Actors

i. **Anna Darpan end User:**
Primary actor utilizing the Ticket Management System to lodge complaints, raise tickets, and seek resolution for issues encountered while interacting with the system or services. These users typically include:
   b. FCI officials
   c. External Users like – Miller, Contractors, Farmers & Agencies, etc.

1.6 References

The following documents and activities have been used as key references for the development of the Ticket Mechanism Module in the Functional Requirement Specification (FRS). These materials provided insights, guidelines, and foundational information for designing the module:

i. **RFP Document for Anna Darpan**
The Request for Proposal (RFP) for the Anna Darpan project provides the requirements, objectives, and expectations for the digitization and streamlining of FCI operations.

ii. **Workshop conducted at Headquarters**
Workshops held at the FCI headquarters played a critical role in shaping the Movement Module by offering a collaborative platform for gathering detailed business requirements.

iii. **Zone-wise Visit happened in August month**
Detailed summaries of visits conducted to various FCI zones and regions, including observations, operational insights, and challenges specific to the movement of commodities within those zones.

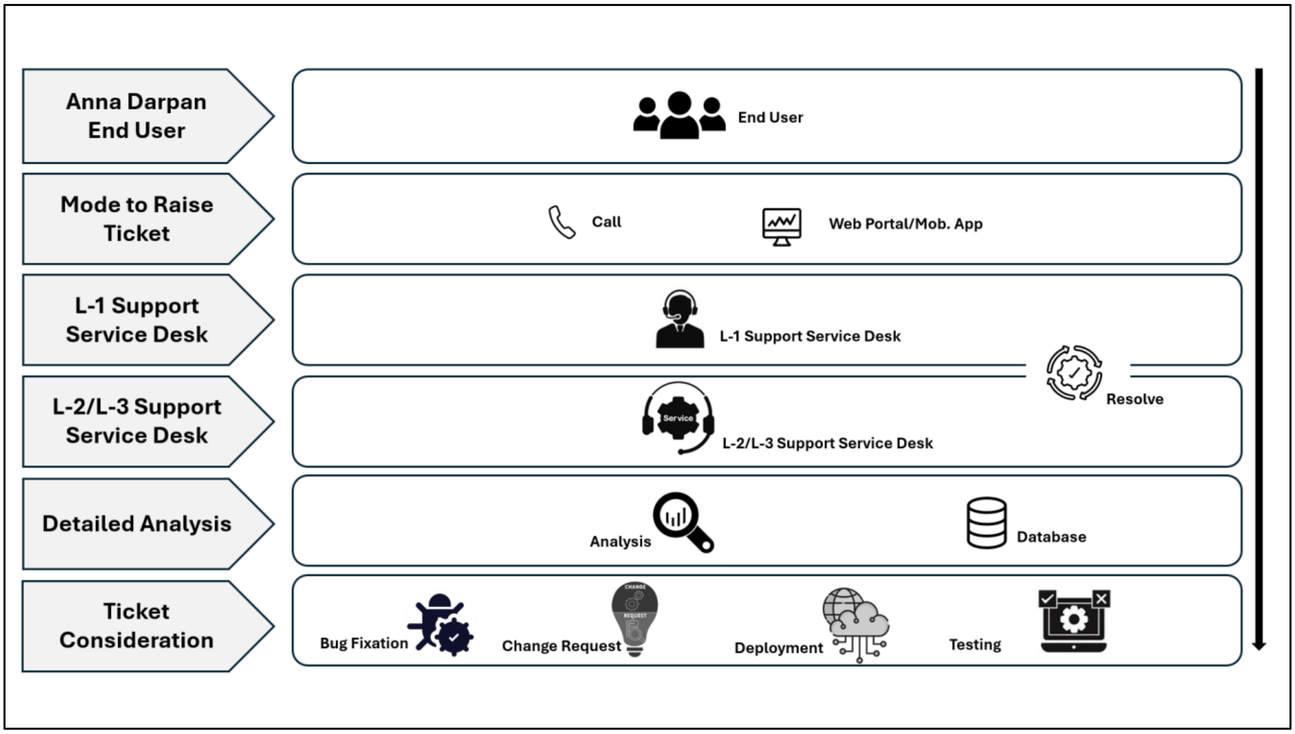
These references serve as foundational materials for the FRS, ensuring that the Ticket Mechanism Module adheres to FCI’s existing processes, legal frameworks, and the operational realities on the ground.

2 Overview of the Module

2.1 Module Description

This Module is designed to streamline and automate the process of managing service requests and incidents raised by Anna Darpan End Users. The system supports multiple levels of support (L1, L2, L3) and enables efficient ticket generation, classification, escalation, resolution, and closure.

The workflow ensures that incidents and service requests are handled in a structured manner, with predefined escalation processes based on issue complexity and resolution failure at earlier support levels. The system ensures timely issue resolution and tracks the performance of each ticket.



Workflow 1: Overview on Ticket Management

2.1.1 Defined Criteria for L1, L2, and L3 Support Levels

i. **L1 Support (Level 1 - Frontline Support):**
   a. **Role:** First point of contact for users/customers.
   b. **Responsibilities:** Basic troubleshooting and resolving issues. Escalates complex issues to L2.
   c. **Examples:** Password resets, basic software troubleshooting, and general information requests.

ii. **L2 Support (Level 2 - Technical Support):**
   a. **Role:** More in-depth troubleshooting and problem-solving.
   b. **Responsibilities:** Handles issues escalated from L1, perform more complex diagnostics, and provide solutions that require deeper technical knowledge.
   c. **Examples:** software minor bugs and malfunctions that L1 cannot resolve.

iii. **L3 Support (Level 3 - Expert/Development Support):**
   a. **Role:** Advanced troubleshooting, system major bug fixes, and code-level debugging.
   b. **Responsibilities:** Handles the most complex issues, often involving code level debugging, and deep technical expertise.
   c. **Examples:** System outages, critical software bugs.

Note: A "complex issue" refers to problems that require more advanced technical knowledge and troubleshooting skills to resolve.

2.2 Key Functional Requirements

i. **Multi-Channel Ticket Logging:**
   a. End users must be able to log service requests or incidents through multiple channels, including web interfaces and mobile applications after login.
   b. End users should also have the option to call/email the Helpdesk, where a representative can record the issue and raise a ticket on their behalf.

ii. **Unique Ticket ID Generation:**
   a. The system should generate unique Ticket IDs for each submitted ticket.
   b. These Ticket IDs should be received by the L1 Lead for further classification and assignments.

iii. **Notifications:**
   a. The system should send automated notifications to users and support staff at various stages of the ticket lifecycle, including ticket creation, status updates, escalations, and resolutions.
   b. Notifications can be sent via email/SMS or through the system's notifications to ensure timely communication.
   c. After a ticket is resolved, the system should trigger an email or notification to end users to record their satisfaction level and provide feedback on the resolution process.

iv. **Tracking of Tickets:**
   a. The system should provide real-time tracking of tickets, allowing end users and support staff to monitor the progress and status of each ticket.
   b. The system should maintain a history of all actions taken on a ticket, including communications.

v. **Ticket Classification and Assignment:**
   a. The L1 Lead, as the first helpdesk support user, is responsible for classifying the severity of the tickets.
   b. Based on the severity, the L1 Lead should assign the tickets to L1, L2, or L3 support users.

vi. **L1 Support User Responsibilities:**
   a. L1 support users are responsible for resolving tickets at the basic troubleshooting level.
   b. They provide initial support and attempt to resolve common issues.
   c. L1 support can mark the ticket as “Assigned to FCI” in case of any dependency lies with FCI services and close the ticket.

vii. **L2 Support Responsibilities:**
   a. L2 support handles tickets escalated by L1 support.
   b. They deal with more complex incidents that require deeper technical knowledge.
   c. L2 support can mark the ticket as "Assigned to FCI" in case of any dependency lies with FCI services and close the ticket.

viii. **L3 Support Responsibilities:**
   a. L3 support manages highly technical and complex incidents that L2 support is unable to resolve.
   b. This level may involve collaboration with developers to resolve the issue.
   c. L3 support can mark the ticket as "Assigned to FCI" in case of any dependency lies with FCI services and close the ticket

ix. **Ticket auto close or manually closed**:
   a. Any ticket marked as "Resolved" should be automatically closed within 72 hours.
   b. Both support users (L1, L2, L3) and end users should have the provision to manually close the ticket.

x. **Change Request Handling:**
   a. Any changes which are supposed to be done in the code level or beyond the agreed requirements should be considered as Change Request.
   b. Tickets that are considered for change requests should be marked as closed with an appropriate description.

xi. **Knowledge base:**
   a. The knowledge base should be a centralized repository of articles, SOPs, FAQs, and troubleshooting tips accessible to L1, L2, and L3 support users.

xii. **Escalation Triggers:**
   a. The system must trigger escalations based on predefined criteria, such as unresolved tickets within SLA limits.
   b. System should have the list of defined users, who should receive the escalation on SLA breach.

SLA/Escalation Matrix

Priority Level	SLA for Resolution	Escalation Level
Critical (P1)	30 minutes	All defined User
High (P2)	1 hour	All defined User

Medium (P3)	4 hours	All defined User
Low (P4)	1 working day	All defined User
Submission of Root Cause Analysis (RCA) reports (For Critical, High and Medium)	5 working days	All defined User

Note: Tickets marked with status as "Assigned to FCI" or "FCI's vendor/partner Third party Dependency", should never be considered for SLA breach. Assigned to FCI should be marked only for those tickets which have any dependency on FCI. These should be closed also along with the above mentioned status.

xiii. Here, user should select the Priority level based on following criteria:

Priority	Description
Critical	1. Entire Application not available 2. Anything that requires a change to be made immediately with the email confirmation consent from FCI. 3. System displaying wrong calculation 4. Workflow not functioning as desired for critical reports 5. Data theft/loss/corruption 6. Any incident escalated by FCI authorized personnel through their login, marking the severity (critical) and the number of users impacted, allows the L1 Lead to freeze the severity with mutual agreement between the SI and FCI.
High	1. Any incident that prevents fifty percent or more users from using the application. 2. Difficult work-around to mitigate the disruption in service 3. Repeat calls (same incident that has occurred earlier and reported more than two times), before the Ticket gets closed 4. Any incident escalated by FCI authorized personnel through their login, marking the severity (High) and the number of users impacted, allows the L1 Lead to freeze the severity with mutual agreement between the SI and FCI.
Medium	1. Any incident that prevents 25% to 50% users from using the application 2. Part of the application not available or not working as desired like system bugs

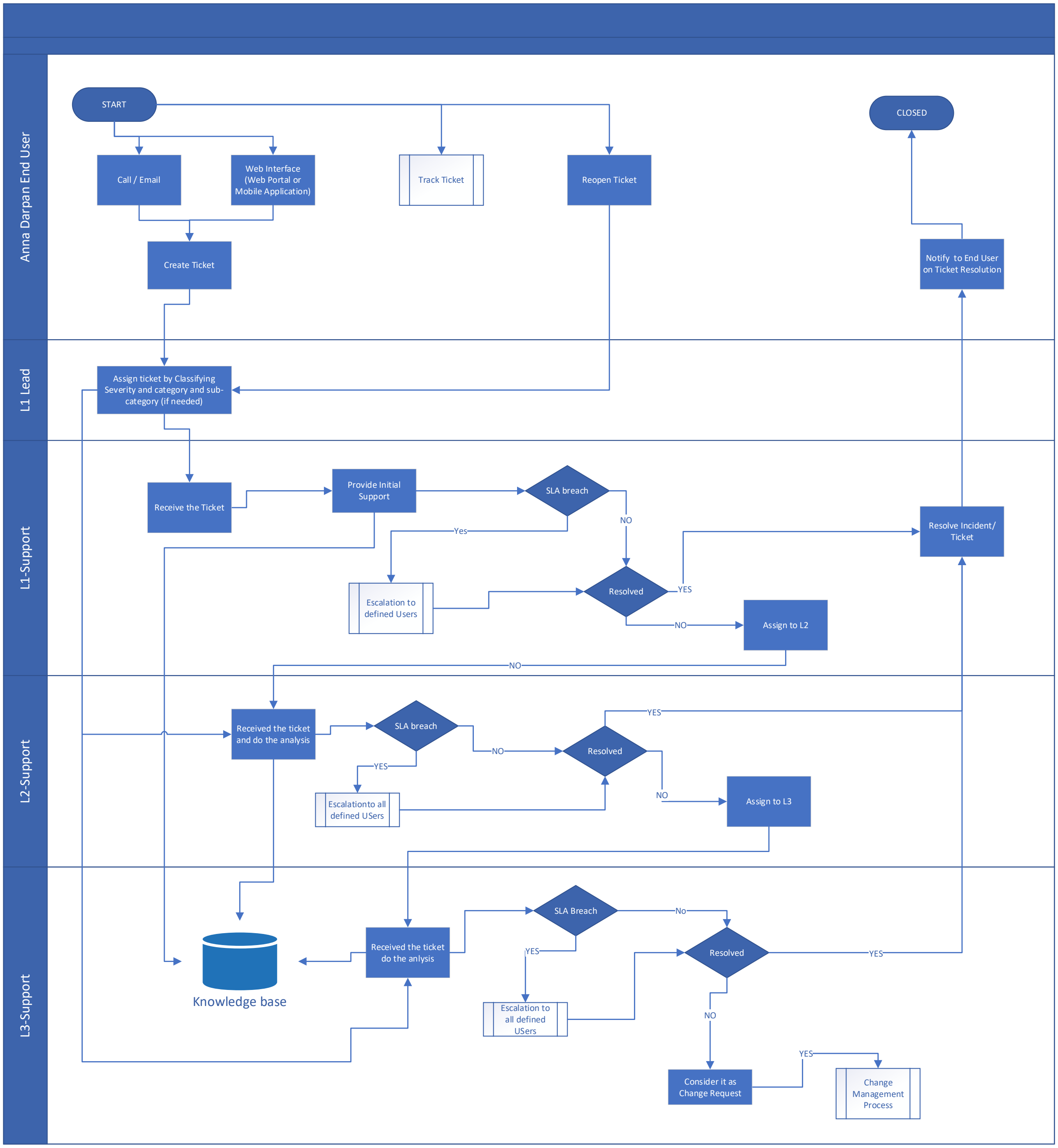
	3. Internal user escalations for slow response of the system impacting the efficiency of users 4. Problem adversely affects product functioning, but a workaround is available
Low	1. A low impact on the efficiency of users 2. No impact on processing of normal business activities. 3. Has a simple workaround 4. Enhancement requests like cosmetic user interface changes, usability improvement, unclear error messages or similar

xiv. **Root Cause Analysis (RCA):**
System should record the Root cause analysis (RCA) against the Critical, High and Medium Tickets with atleast following fields:
a. Brief Description on issue
b. Impact to productivity (High, Medium or Low)
c. Potential reason, why the issue happened
d. Describe the resolution details
e. Mitigation and Permanent fix
f. Attachment (if any)

xv. **SLA start timing:**
a. For any newly submitted tickets, the SLA time will be calculated from the submission time.
b. For any tickets escalated by FCI, the SLA time will be calculated from the severity determined/freezed by the L1 Lead as per mutual agreement between SI & FCI.
c. SLA would never be calculated for those tickets which were marked with status as “Assigned to FCI” and “FCI’s vendor/partner Third party Dependency”.

2.3 Workflow

Ticket Management



Workflow 2: Ticket Management

i. The Anna Darpan End User can raise a ticket through either a call/email or the web interface/mobile application, depending on their preferred method for submitting a support request.

ii. Upon receiving the ticket, the L1 Lead assigns the ticket to L1, L2, or L3 support based on the severity and nature of the issue.
iii. The L1 support staff reviews the issue and attempts to resolve it by providing initial support.
iv. If the issue is resolved, the ticket is marked as resolved. If not, the L1 support staff assigns the ticket to L2 support for further handling.
v. System should escalate to defined users in case of SLA breach.
vi. L1, L2, and L3 support staff have access to the Knowledge Base, which contains articles, SOPs, FAQs, and troubleshooting tips to assist in ticket resolution.
vii. Upon receiving the ticket, the L2 support staff updates the incident in the system and accesses the Knowledge Base for potential solutions.
viii. If L2 does not resolve the issue within the defined time frame, the system automatically initiates the escalation process to defined users.
ix. If L2 identifies that the issue requires advanced expertise or further technical support, they assign the ticket to L3 support staff after conducting a thorough workaround of the incident to gather all necessary details.
x. The L3 support staff either resolves the issue (e.g., bug fixes, advanced troubleshooting) or may consider the incident as a Change Request if the issue requires changes in the code level.
xi. The Change Request process involves a complete offline process, including separate documentation with impact analysis and implementation plans.

2.4 Role Based Access

i. **Anna Darpan End User:**
   a. All Anna Darpan end users should have the basic privilege to create and track tickets. This includes submitting new tickets for support requests and tracking the status of those tickets until resolution.

ii. **L1 Lead:**
   a. The L1 Lead is responsible for classifying the severity of incoming tickets and assigning them to the appropriate support level (L1, L2, or L3).
   b. Responsible for creating the Master Ticket and manually link the other tickets having same issue with it.

iii. **L1 Support Staff:**
   a. L1 support staff are typically the first point of contact for incoming tickets. They handle the initial review and attempt to resolve functional issues.
   b. L1 support staff can assign tickets to L2 support if the issue cannot be resolved at their level.

iv. **L2 Support Staff:**
   a. L2 support handles more complex issues that cannot be resolved by L1 support.
   b. L2 support staff have access to the Knowledge Base to assist in resolving tickets.
   c. L2 support staff can escalate tickets to L3 support if further expertise is required.

v. **L3 Support Staff**:
   a. L3 support typically consists of highly skilled experts. They are responsible for handling the most complex, critical issues that cannot be resolved by L1 or L2 support.
   b. L3 support staff can identify and manage incidents that require a Change Request process.

Note: Roles and permissions are managed through the User Management module, so the specific rights as per role are not detailed above.

2.5 Dependencies with Other Modules

The Ticket Mechanism within the Anna Darpan is intricately linked with Masters and MIS Reporting modules.

i. **Masters:** Masters would be supporting Ticket Mechanism module in getting the category master, sub-category master and also the Escalation defined user list.
ii. **MIS Reporting:** MIS Reporting module would require access to the consolidated data of tickets raised, resolved, average time, etc.

2.6 MIS Reports

As per the RFP, the MIS Report is defined as a separate module, and therefore, following reports are covered:

i. **Ticket Summary Report:** To provide an overview of all tickets submitted to the helpdesk by users of ANNA DARPAN including details like ticket counts, ticket status, mode of receipt or similar details.
ii. **Ticket Resolution Time Report:** To provide details of the average resolution time taken for resolving tickets.
iii. **Customer Satisfaction Report:** To monitor user feedback and satisfaction levels with the resolution provided by the helpdesk.
iv. **Problem Management Report:** To track details of category and Sub-category wise type of issues reported for identification of recurring issues.

Reports are covered in the MIS Reports module FRS, and concerned divisions of FCI shall provide MIS Reports templates covering maximum fields/parameters for generation of static and dynamic straight reports.

3 Sub-Modules

3.1 Ticket Creation

3.1.1 Description

The system must allow end-users to create tickets via two primary channels:

i. **Call/email:**
   a. **Helpdesk Interaction:** End users can call on the Toll free number or email on the Helpdesk email id present on the Web page (pre-login) to report an issue. The Helpdesk representative would capture the details of the issue and generate a ticket on behalf of the user.
   b. **User Identification:**
      I. Before capturing the issue, Helpdesk users should ask for the registered Email id, Mobile no. or Employee ID (in case of FCI officials).
      II. System should validate the user and fetch the basic information like Name, Email, Mobile No. and may also fetch the role and office detail (in case of FCI officials).
   c. **Call Recording:** System record the call and should retained for a specified period before being automatically refreshed or deleted to manage storage efficiently.
   d. **Ticket Generation:** The Helpdesk representative would enter the issue details into the system, which would then generate a ticket ID.

ii. **Web Interface:**
   a. **Web Portal/Mobile Application:** End users can log in to the Anna Darpan web portal or mobile application to create a ticket.
   b. **User Identification:** System should fetch the basic detail of the user as per the login basis and display on the screen like Name, Email, Mobile No. and may also fetch the role and office detail (in case of FCI officials).
   c. **User Input:** Users should fill out a form detailing the issue they are experiencing and submit it through the web/mobile interface.

3.1.2 Core Functionalities

i. **Auto-Generation of Ticket ID:**
   a. **Unique ID:** Upon ticket creation, the system would automatically generate a unique ticket ID for each new ticket. This ID would be used for tracking and reference purposes.

ii. **Automated Workflow:**
   a. **L1 Level Assignment:** The system would automate the workflow to ensure that the newly created ticket is routed to the L1 support level for initial review and resolution.

iii. **Notification Triggers:**
   a. **Email/SMS Notifications:** The system would trigger notifications via email or SMS to inform the end user that their ticket has been successfully created. This notification would include the unique ticket ID and other relevant details.
   b. **System notification:** As soon as the ticket gets submitted, system should trigger the notification to L1 Lead.

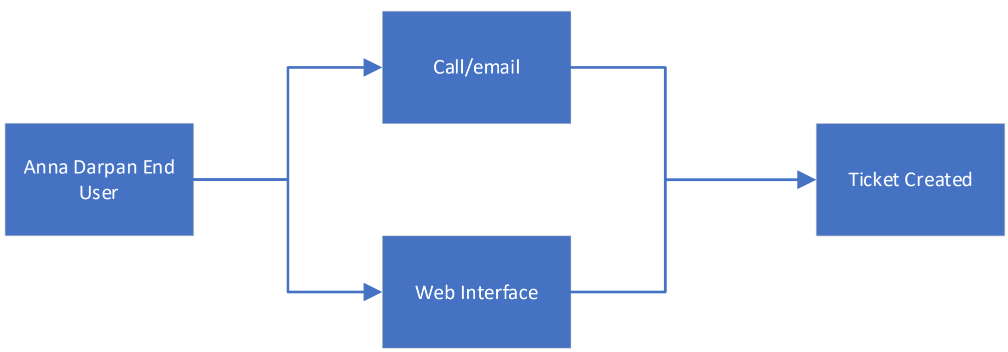
iv. **Handling Login Issues:**

a. **Helpdesk Assistance:** If an end user encounters login issues and is unable to access the web portal or mobile application, they must call/email the Helpdesk for assistance.
b. **Ticket Creation by Helpdesk:** The Helpdesk representative should raise a ticket on behalf of the user, capturing the details of the login issue.
c. **Toll-Free Number and helpdesk email ID:** A toll-free number and helpdesk email ID should be prominently displayed on the Anna Darpan web page (before login/home page) to ensure users know how to contact the Helpdesk for support.

v. **Ticket Creation Form:**
a. **Ticket Categorization:** End users need to select the category, sub-category, and priority of the ticket during the creation process. This helps in accurately classifying and prioritizing the issue.
b. **Brief Description:** End users should have the provision to write a brief description of the issue they are experiencing.
c. **Attachments:** Users should also have the option to attach relevant documents, such as PDFs or screenshots, to provide additional context and information about the issue.

3.1.3 Workflows

Create Ticket



Workflow 3: Create Ticket

i. Anna Darpan End Users would be having the provision to lodge Complaint either by calling/email on the Toll-free number/helpdesk email displayed on the Web Portal or by raising the complaint through Helpdesk section.
ii. System should generate the unique Ticket No.

3.1.4 Role-Based Access

i. **Anna Darpan End User:** Any user who has the login credential in Anna Darpan would have the provision to create the ticket, specifically to Anna Darpan only.

ii. **Helpdesk:** Officials/staff from Helpdesk team, who would be supporting Anna Darpan End users queries over the call. They would be supporting in creating the ticket too.

3.2 L1 Lead (Ticket acknowledgement and assignment)

3.2.1 Description

The L1 Lead plays a crucial role in the ticket management process by overseeing the initial classification and assignment of tickets. L1 Lead ensure that tickets are routed to the appropriate support level (L1, L2, or L3) based on the severity and nature of the issue. Additionally, the L1 Lead can identify and manage duplicate tickets by marking a initial ticket as the Master Ticket and linking related tickets to it.

3.2.2 Core Functionality

i. **Ticket Classification and Assignment:**
   a. **Severity Assessment:** The L1 Lead assesses the severity of incoming tickets and classifies them accordingly.
   b. **Assignment:** Based on the classification, the L1 Lead assigns tickets to the appropriate support level (L1, L2, or L3).
   c. **System Notification:** System should trigger the notification to concern L1 official as soon as L1 Lead assign the tickets.

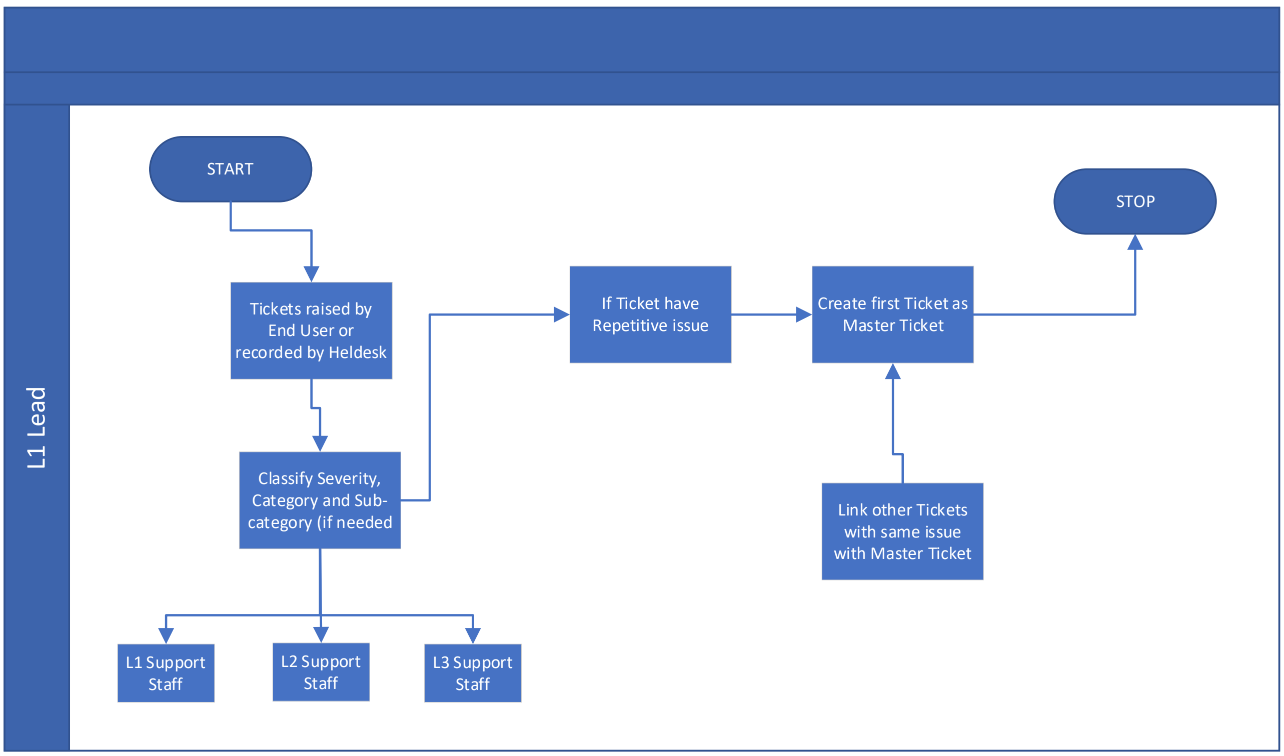
ii. **Master Ticket Management:**
   a. **Duplicate Ticket Identification:** If the L1 Lead identifies consistent ticket creation for the same issue, they can mark the first ticket as the Master Ticket.
   b. **Linking Tickets:** The L1 Lead closes all other related tickets by linking them to the Master Ticket. Also assign the Severity of the Master ticket based on the total user effected.
   c. **Tracking:** End users are informed to track the incident through the Master Ticket ID through email/SMS.

iii. **Category, Sub-Category, and Severity Adjustment:**
   a. **Provision for Changes:** The L1 Lead has the provision to change the category, sub-category, and select the severity of the ticket along with adding remarks.
   b. **SLA Applicable:** The SLA would apply based on the severity selected by the L1 Lead. For example, if an end user selects "Critical" as the priority but the L1 Lead determines it is a "Medium" severity ticket, the SLA would apply as per the medium severity. However, the SLA time would count from the ticket submission time by the end user.
   c. **Logging Changes:** The system maintains a proper log of any changes made to the category, sub-category, or severity of the ticket by L1 Lead.
   d. **Exceptional circumstances:** If FCI emails and confirms the severity marking for any specific ticket, L1 Lead should update the ticket's severity accordingly.

iv. **Freezing the Severity of ticket for escalated Tickets with mutual agreement:**

a. The FCI regional nodal officer will have the authority to escalate tickets with a recommended severity level, ensuring quicker resolution. These escalated tickets will be reviewed and finalized by the IT division (any concerned official) with the final severity and appropriate remarks.
b. The L1 Lead should acknowledge or assign these tickets to the L1 Support Staff.
c. SLA will commence once the L1 Lead receives the escalated ticket.

3.2.3 Workflow



Workflow 4: Work performed by L1 Lead

i. **Ticket Reception:**
   a. The L1 Lead receives incoming tickets from end users via the Helpdesk or web interface/mobile application.

ii. **Classification and Assignment:**
   a. The L1 Lead assesses the severity of each ticket and assigns it to the appropriate support level (L1, L2, or L3).

iii. **Master Ticket Management:**
   a. Upon identifying duplicate tickets for the same issue, the L1 Lead marks the first ticket as the Master Ticket.

b. The L1 Lead closes all related tickets and links them to the Master Ticket.
c. Also assign the Severity of the Master ticket based on the total user effected.
d. End users are notified to track the incident through the Master Ticket ID.

iv. **Category, Sub-Category, and Severity Adjustment:**

a. The L1 Lead can change the category, sub-category, and severity of tickets, with the system maintaining a log of these changes

3.2.4 Role Based Access

i. L1 Lead

a. The L1 Lead has the authority to classify the severity of tickets and assign them to the appropriate support level (L1, L2, or L3)
b. The L1 Lead can identify duplicate tickets, mark a primary ticket as the Master Ticket, and link related tickets to it. Also assign the Severity of the Master ticket based on the total user effect.
c. Change the Category, Sub-category and Severity of the Ticket (if required).

3.3 L1 Support: Initial Ticket Handling

3.3.1 Description

L1 Support Staff are the first point of contact for end users who have raised tickets. They handle the initial review and attempt to resolve functional issues. Their primary responsibility is to provide basic troubleshooting and support, ensuring that issues are resolved quickly and efficiently. If the issue is beyond their scope, they escalate the ticket to L2 support.

3.3.2 Core Functionalities

i. **Initial Review and Resolution:**

a. **Issue Assessment:** L1 Support Staff review the details of the ticket to understand the issue.
b. **Basic Troubleshooting:** They perform basic troubleshooting steps to resolve common issues.

ii. **Ticket Assignment**

a. **Escalation:** If the issue cannot be resolved at the L1 level, the support staff escalate the ticket to L2 support.

iii. **Knowledge base Access**

a. **Utilization:** L1 Support Staff have access to the Knowledge Base to find solutions and troubleshooting steps for common issues.

iv. **Notification:**

a. Once the ticket is resolved, system trigger alert/notification to end user confirming the Ticket is successfully resolved.

3.3.3 Workflow

i. Ticket Review and workaround:

a. L1 Support Staff receive tickets assigned by the L1 Lead or directly from end users via the Helpdesk or web interface/mobile application.
b. They review the ticket details to understand the issue and determine the appropriate troubleshooting steps.
c. L1 Support Staff perform basic troubleshooting to resolve common issues. They use the Knowledge Base to assist in finding solutions.
d. If the issue cannot be resolved at the L1 level, they escalate the ticket to L2 support or involve external vendors if necessary.

Workflow is shown in Workflow -2 under section 2.3

3.3.4 Role–based Access

i. L1 support official/staff

a. L1 Support Staff have access to review and attempt to resolve tickets assigned to them.
b. They have the authority to escalate tickets to L2 support or involve external vendors if the issue cannot be resolved at their level.
c. L1 Support Staff have full access to the Knowledge Base to assist in troubleshooting and resolving issues.

3.4 L2 Support: Advanced Incident Management

3.4.1 Description

L2 Support Staff handle more complex issues that cannot be resolved by L1 support. They provide in-depth technical support and troubleshooting for escalated tickets. Their role involves analyzing and resolving issues that require a higher level of expertise. L2 Support Staff also collaborate with L3 support and external vendors when necessary to ensure the resolution of complex incidents.

3.4.2 Core Functionalities

i. **Advanced Troubleshooting and Resolution:**

a. Issue Analysis: L2 Support Staff analyze the details of escalated tickets to understand the root cause of the issue.
b. Technical Support: They provide advanced technical support and troubleshooting to resolve complex issues.

ii. **Ticket Assignment and Escalation:**

a. **Further Escalation:** If the issue cannot be resolved at the L2 level, the support staff escalate the ticket to L3 support.
b. **Vendor Collaboration:** L2 Support Staff can collaborate with external vendors if the issue requires specialized expertise.

iii. **Knowledge Base Access:**
a. **Utilization:** L2 Support Staff have access to the Knowledge Base to find solutions and troubleshooting steps for complex issues.

iv. Notification:
a. Once the ticket is resolved, system trigger alert/notification to end user confirming the Ticket is successfully resolved.

3.4.3 Workflow

i. Ticket Review and workaround:
a. L2 Support Staff receive tickets escalated by L1 support or directly assigned by the L1 Lead.
b. They analyze the ticket details to understand the root cause of the issue and determine the appropriate troubleshooting steps.
c. L2 Support Staff perform advanced troubleshooting to resolve complex issues. They use the Knowledge Base to assist in finding solutions.
d. If the issue cannot be resolved at the L2 level, they escalate the ticket to L3 support or involve external vendors if necessary.

Workflow is shown in Workflow -2 under section 2.3

3.4.4 Role – based Access

i. L2 support staff
a. L2 Support Staff have access to review and attempt to resolve tickets assigned to them.
b. They have the authority to escalate tickets to L3 support or involve external vendors if the issue cannot be resolved at their level.
c. L2 Support Staff have full access to the Knowledge Base to assist in troubleshooting and resolving issues.

3.5 L3 Support: Expert Troubleshooting and Resolution

3.5.1 Description

L3 Support Staff are highly skilled experts responsible for handling the most complex and critical issues that cannot be resolved by L1 or L2 support. They provide advanced technical support, including system fixes and code-level debugging. L3 Support Staff play a crucial role in ensuring the resolution of critical incidents and may also manage change requests that require modifications to the system.

3.5.2 Core Functionalities

i. **Advanced Issue Resolution**:

a. **Technical Expertise:** They provide advanced technical support, including system fixes, code-level debugging, and performance tuning.

b. **Bug fixation:** L3 resolve the bugs related to the Anna Darpan application.

c. **Utilization:** L3 Support Staff have access to the Knowledge Base to find solutions and troubleshooting steps for highly technical issues.

ii. Notification:

a. Once the ticket is resolved, system trigger alert/notification to end user confirming the Ticket is successfully resolved.

3.5.3 Workflow

i. Ticket Review and workaround:

a. L3 Support Staff receive tickets escalated by L2 support or directly assigned by the L1 Lead.

b. L3 Support Staff perform advanced troubleshooting to resolve complex issues. They use the Knowledge Base to assist in finding solutions.

c. If the issue requires modifications to the system, L3 Support Staff manage the change request process, including documentation and impact analysis.

Workflow is shown in Workflow -2 under section 2.3

3.5.4 Role-based Access

i. L3 Support Staff

a. L3 Support Staff have access to review and attempt to resolve tickets assigned to them.

b. L3 Support Staff have full access to the Knowledge Base to assist in troubleshooting and resolving issues.

c. Tickets which are marked for change request is marked as closed and notified to the End user.

3.6 Knowledge base

3.6.1 Description

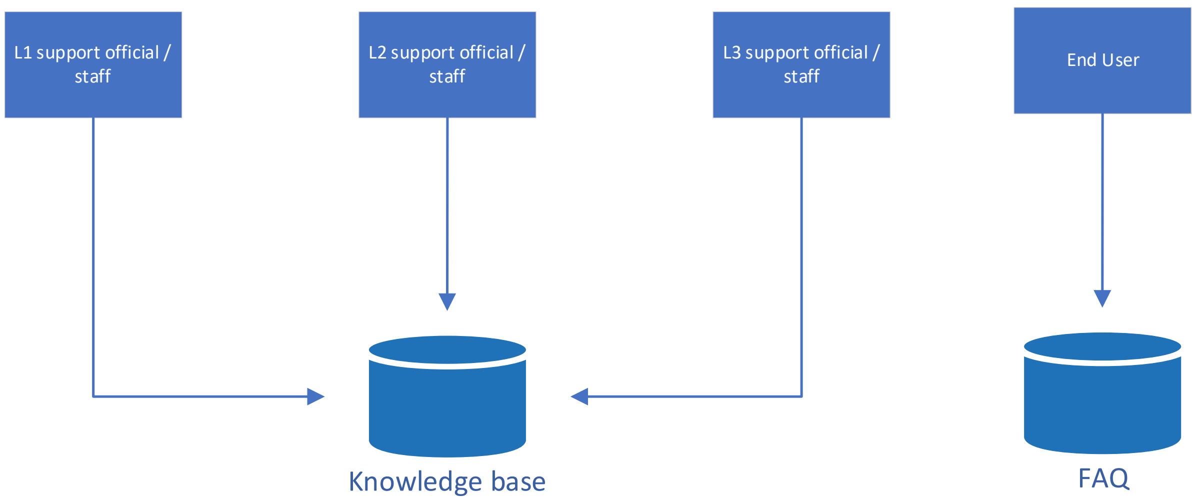
The Knowledge Base is a centralized repository of information designed to assist support staff at all levels (L1, L2, and L3) in resolving tickets efficiently. It contains articles, guides, SOPs, FAQs, troubleshooting tips, and best practices. The Knowledge Base is regularly updated to ensure it remains a valuable resource for support staff, helping them find solutions to common and complex issues.

3.6.2 Core Functionalities

i. **Repository**:

a. The Knowledge Base includes a wide range of information, such as troubleshooting guides, SOPs, FAQs, and User Manuals.
b. Information is categorized by topic, description, and attachment (if required).
c. Search functionality allows support staff to quickly find relevant articles and information by entering keywords or phrases.
d. Knowledge base should be configurable through UI.
e. System administrator or any specific FCI official having its right should able to manage it CRUD (Create/Read/Update/Deactivate) operation through UI.

3.6.3 Workflow



L1 Lead, L1, L2 and L3 support staff should be able to access the knowledge base whereas End User would be having the access to FAQ

3.6.4 Role-based Access

i. L1 support official/staff
L1 Support Staff have full access to the Knowledge Base to assist in resolving common issues
ii. L2 support official/staff
L2 Support Staff have full access to the Knowledge Base to assist in resolving more complex issues.
iii. L3 support official/staff
L3 Support Staff have full access to the Knowledge Base to assist in resolving the most complex issues.
iv. End User
End User would be having access to only FAQ section.

3.7 Ticket Tracking, Reopening and record the Satisfactory level against the ticket by Anna Darpan End User

3.7.1 Description

The ticket tracking and reopening functionalities allow end users to monitor the status of their submitted tickets and reopen them if the issue persists or if the resolution provided is unsatisfactory. These features ensure that end users remain informed about the progress of their tickets and have the ability to seek further assistance if needed.

System should also always trigger a mail or notification to record the satisfaction level after any Ticket resolution.

3.7.2 Core Functionalities

i. **Ticket Tracking:**
   a. **Real-Time Updates:** End users can track the status of their tickets in real-time through the web portal or mobile application.
   b. **Status Notifications:** The system sends automated notifications to end users about significant status changes, such as ticket assignment, progress updates, and resolution.
   c. **Suggested Status:**
      I. Work in progress: The ticket is submitted and currently being worked on by the support team
      II. Assigned to Support user: The ticket has been assigned to a specific support user for resolution
      III. Resolved: The issue has been resolved, and the ticket is awaiting confirmation from the end user
      IV. Closed: The ticket has been closed after resolution or due to no further action required
      V. Re-open and submitted: The ticket has been reopened by the end user and resubmitted for further investigation.

ii. **Reopening Tickets:**
   a. Reopening Criteria: End users can reopen tickets if the issue reoccurs or if they are not satisfied with the resolution provided.
   b. Reopening Process: The system allows end users to reopen tickets through the same channels used for ticket creation, with an option to provide additional details or feedback.
   c. Timeline: A ticket can be reopened within 72 hours of its resolution. If a ticket is not reopened within this timeframe, the system should automatically close the ticket.

iii. **Satisfaction Level Recording:**
   a. Feedback Request: The system should always trigger an email or notification to end users to record their satisfaction level after any ticket resolution.
   b. Feedback Collection: End users can provide feedback on the resolution process, which helps in improving the support services.
   c. Sample form (customer satisfaction form):

Customer Satisfaction Form

1. **Ticket Information**
   Ticket ID: [Auto-filled]
   Date of Resolution: [Auto-filled]

2. Satisfaction Rating
Please rate your overall satisfaction with the resolution provided:
⭐⭐⭐⭐⭐ Very Satisfied
⭐⭐⭐⭐ Satisfied
⭐⭐⭐ Neutral
⭐⭐ Dissatisfied
⭐ Very Dissatisfied

3. Resolution Effectiveness
How effective was the resolution provided by the helpdesk?
⭐⭐⭐⭐⭐ Very Effective
⭐⭐⭐⭐ Effective
⭐⭐⭐ Neutral
⭐⭐ Ineffective
⭐ Very Ineffective

4. Communication and Support
How would you rate the communication and support provided by the helpdesk staff?
⭐⭐⭐⭐⭐ Excellent
⭐⭐⭐⭐ Good
⭐⭐⭐ Neutral
⭐⭐ Poor
⭐ Very Poor

5. Timeliness
How satisfied are you with the timeliness of the resolution?
⭐⭐⭐⭐⭐ Very Satisfied
⭐⭐⭐⭐ Satisfied
⭐⭐⭐ Neutral
⭐⭐ Dissatisfied
⭐ Very Dissatisfied

6. Additional Comments/Feedback
Please provide any additional comments or feedback to help us improve our support services:
Comments: [Text Box]

Submit **Cancel**

3.7.3 Workflow

i. **Ticket Tracking:**

a. **Access:** End users log in to the Anna Darpan web portal or mobile application to track the status of their tickets.
b. **Real-Time Updates:** The system provides real-time updates on the status of the ticket, including assignment, progress, and resolution.
c. **Notifications:** End users receive automated notifications about significant status changes via email or SMS.

ii. **Reopening Tickets:**

a. **Reopening Request:** If the issue reoccurs or the resolution is unsatisfactory, end users can request to reopen the ticket through the web portal or mobile application.
b. **Notification:** The system sends notifications to the relevant support staff to ensure prompt attention to the reopened ticket.
c. **Time Restriction:** End- User can re-open the ticket within 72 hours of its resolution only. After this time period, system should disable the functionality to re-open the ticket.

iii. **Satisfaction Level Recording:**

a. **Feedback Request:** After a ticket is resolved, the system triggers an email or notification to the end user to record their satisfaction level.
b. **Feedback Collection:** End users provide feedback on the resolution process, which is used to improve support services.

Workflow is shown in Workflow -2 under section 2.3

3.7.4 Role-based Access

i. Anna Darpan End User

a. **Ticket Creation and Tracking:** End users have the basic privilege to create and track tickets through the web portal or mobile application.
b. **Reopening Tickets:** They can reopen tickets if the issue persists or if the resolution provided is unsatisfactory.
c. **Satisfaction Feedback:** End users can provide feedback on their satisfaction level after ticket resolution.

ii. L1 lead, L1, L2 & L3

a. **Notification and Response:** Support staff receive notifications when a ticket is reopened and are responsible for addressing the reopened ticket promptly.

3.8 Root Cause Analysis (RCA)

3.8.1 Description

Root Cause Analysis (RCA) is a systematic process used to identify the underlying causes of problems or incidents. By conducting RCA, support teams can ensure the prevention of recurring issues and validate that the recommended actions can improve processes. RCA is essential for understanding the root causes of critical, high, and medium severity tickets, leading to more effective and long-term solutions.

3.8.2 Core Functionalities

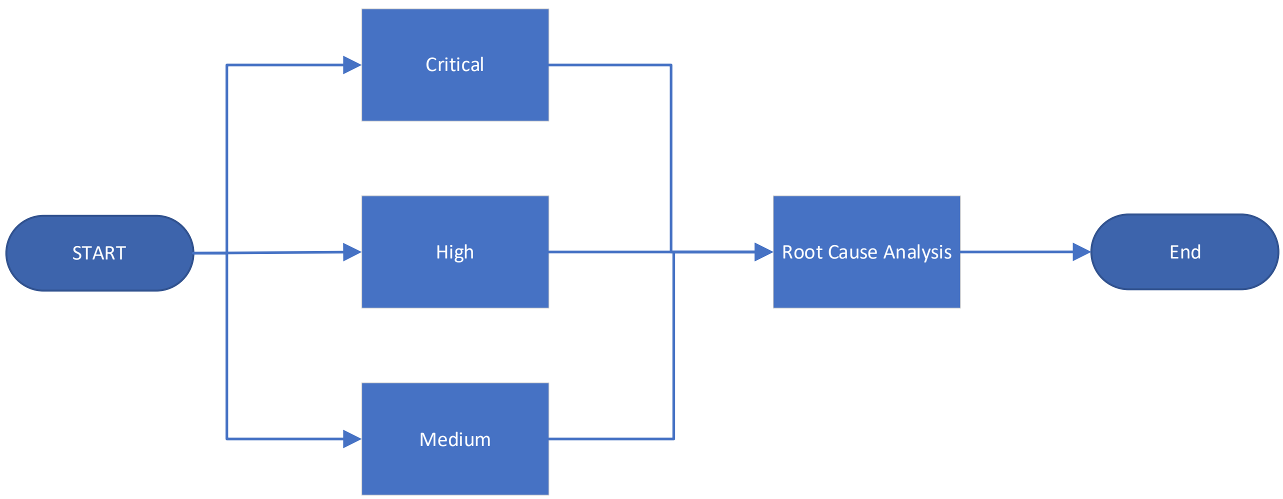
i. **Eligibility for RCA:**
   a. **Ticket Criteria:** Every critical, high, and medium severity ticket from users should be eligible for RCA. This ensures that significant issues are thoroughly investigated to prevent recurrence.

ii. **RCA Entry within SLA:**
   a. **Timely Analysis:** The concerned resolver support user must enter the RCA details against the tickets within the defined SLA. This ensures that the analysis is conducted promptly and does not delay the resolution process.

iii. **Detailed Descriptions and Attachments:**
   a. **Comprehensive Documentation:** Support users need to enter detailed descriptions of the root cause and the steps taken to resolve the issue. They may also attach any supporting documents, such as screenshots, to provide additional context and attachments.

3.8.3 Workflow

Root cause Analysis



Workflow 5: Root Cause Analysis

i. **Ticket Identification:**
   a. **Severity Assessment:** Critical, high, and medium severity tickets are identified as eligible for RCA based on their impact and complexity.

ii. **RCA Initiation:**
   a. **Assignment:** The concerned resolver support user is assigned to conduct the RCA for the identified ticket.
   b. **SLA Compliance:** The support user ensures that the RCA is completed within the defined SLA to maintain timely resolution.

iii. **Root Cause Analysis:**
   a. **Investigation:** The support user conducts a thorough investigation to identify the root cause of the issue. This may involve analyzing system logs, reviewing configurations, and consulting with other team members or external vendors.
   b. **Documentation:** The support user enters detailed descriptions of the root cause and the steps taken to resolve the issue in the system. They may also attach supporting documents to provide additional context.

3.8.4 Role Based Access

i. L1 Support staff
   a. **RCA Participation:** L1 Support Staff may assist in gathering initial information and evidence for the RCA but are not primarily responsible for conducting the analysis.

ii. **L2 Support Staff**
   a. **RCA Execution:** L2 Support Staff are responsible for conducting RCA for high and medium severity tickets. They analyze the root cause and document the findings in the system.

iii. **L3 Support Staff**

a. **Advanced RCA:** L3 Support Staff handle RCA for critical severity tickets and complex issues that require advanced technical expertise. They collaborate with development teams and external vendors if necessary.

3.9 Configurable Category & Sub-category Master

3.9.1 Description

The Configurable Category & Sub-category Master allows the system administrator to manage and customize the categories and sub-categories used for ticket classification. This functionality ensures that the ticketing system remains flexible and can adapt to changing requirements and organizational needs. The categories and sub-categories help in accurately classifying and prioritizing tickets, making it easier for support staff to manage and resolve issues efficiently.

Suggested Master for Category, Sub-category are as follows:

S. No.	Category	Sub-Category	Suggested Severity
1	**System Issues**	Entire application not available	Critical
		SMS gateway not working	High
		Mail gateway not working	High
		System Slow response	Medium
2	**Single User Access**	Login Issues	Low
		Mail not received while resetting Password	Low
		OTP not received during login	Low
		Account Locked	Low
		Role-based access (Role assignment issue)	Low
3	**Data Management**	Data Entry Errors	Low
		Data theft/loss/corruption	Critical
		Data Synchronization	High
4	**Mobile Application related issues**	Mobile android application downloading issue	Medium
		Mobile IOS application downloading issue	Medium
		Unable to login in Mobile application	Low
5	**System bug (may define Module wise)**	Unable to generate Token	Medium
		Unable to generate Gatepass	Medium
		Weighbridge is integrated but unable to record the weight	Medium
		Unable to record quality parameter	Medium
		Unable to record no. of bags/weight (during unloading)	Medium
		6Unable to record no. of bags/weight (during loading)	Medium

		System displaying wrong calculations	Critical
		Unable to generate WCM	Medium
		Unable to generate Truckchit	Medium
		Unable to generate Release Order (RO)	Medium
		Not able to view Release Order (RO)	Medium
		Unable to view or download the report	Medium
6	**Workflow related issue**	workflows not functioning as desired for critical reports	Critical
7	**Cosmetic Changes**	Error message is not clear	Low
		usability improvement	Low
		Minor change on User Interace (UI)	Low
8	**Any other issue related to Anna Darpan only**	<type by user>	Low

Note: Above master is only the indicative. Since master is configurable, it may be configured by System Administration through UI with mutual discussion with FCI and SI. Also, severity is selected by L1 Lead based on the defined criteria.

3.9.2 Core Functionalities

i. **Category and Sub-category Management:**
   - **a. Creation:** System administrators can create new categories and sub-categories to reflect the evolving needs of the organization.
   - **b. Modification:** Existing categories and sub-categories can be modified to ensure they remain relevant and accurate.
   - **c. Deletion:** Outdated or redundant categories and sub-categories can be deleted to maintain a clean and organized system.
   - **d.** The system provides an intuitive user interface for system administrators to manage categories and sub-categories easily.

3.9.3 Workflow

i. **Accessing the Category & Sub-category Master:**
   - **a. Administrator Login:** System administrators log in to the Anna Darpan system with their credentials.
   - **b. Navigation:** They navigate to the Category & Sub-category Master section through the system's user interface.

ii. **Creating New Categories and Sub-categories:**
   - **a. New Entry:** Administrators can create new categories and sub-categories by entering the relevant details in the provided fields.
   - **b. Save Changes:** Once the details are entered, administrators save the changes to add the new categories and sub-categories to the system.

iii. **Modifying Existing Categories and Sub-categories:**

a. **Selection:** Administrators select the category or sub-category they wish to modify.
b. **Edit Details:** They update the necessary details and save the changes to reflect the modifications in the system.

iv. **De-activating Categories and Sub-categories:**
a. **Selection:** Administrators select the category or sub-category they wish to delete.
b. **Confirmation:** They confirm the deletion to remove the selected category or sub-category from the system.

3.9.4 Role Based Access

i. System Administrators:
a. **Full Access:** System administrators have full access to manage categories and sub-categories through the user interface.
b. **Creation, Modification, and Deletion:** They can create, modify, and delete categories and sub-categories as needed.

3.10 Regional Nodal Officer to escalate any specific Ticket (if required)

3.10.1 Description

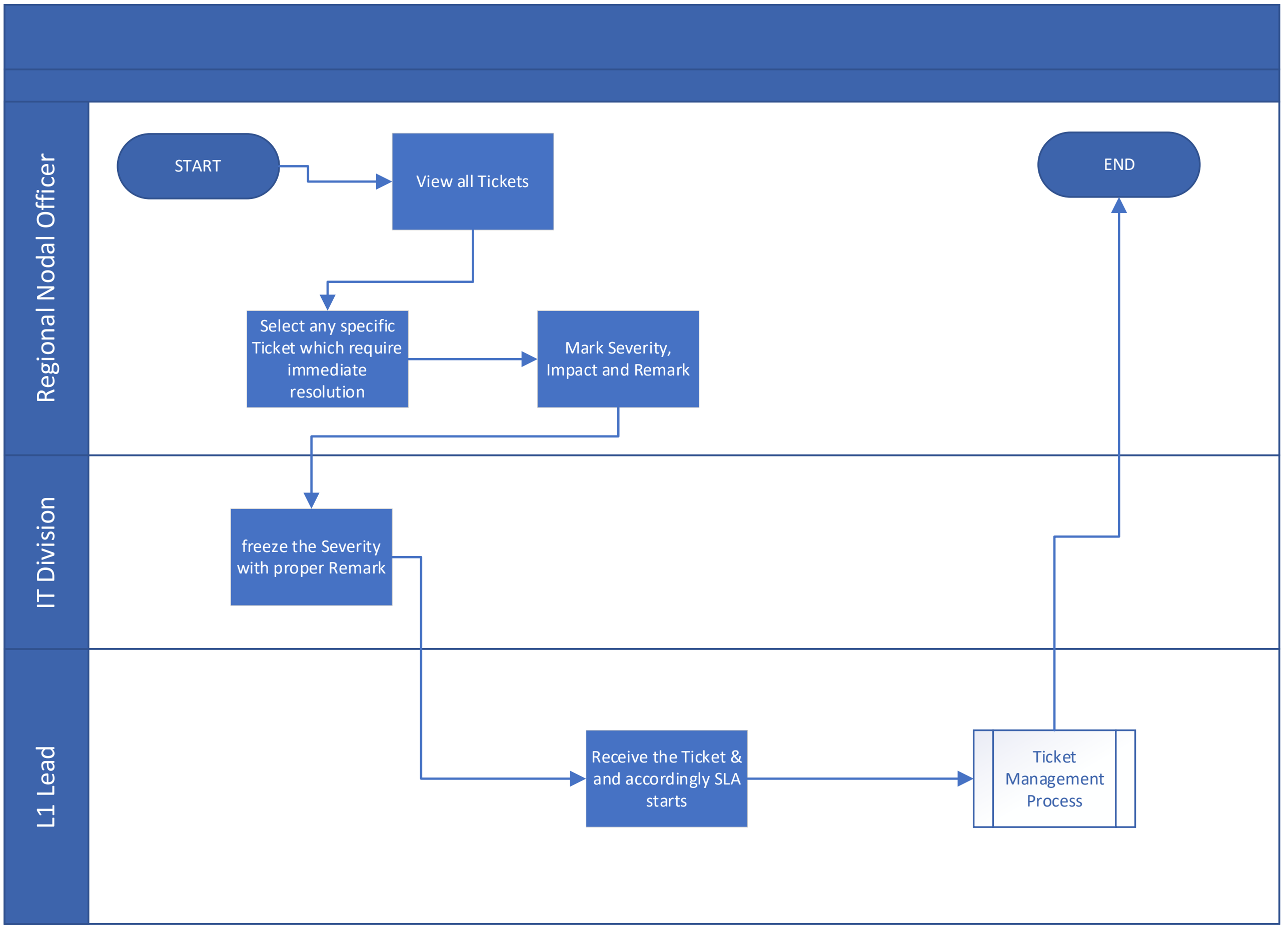
The system is designed to manage and escalate support tickets within the Food Corporation of India (FCI), if needed. It ensures that tickets are efficiently handled and escalated based on their severity and impact on users. In that case, the system involves multiple roles, including Nodal Officers, any concern official from IT Division and L1 Leads, to streamline the ticket management process.

i. This process is only for specific tickets which require higher priority on resolution.
ii. SLA for these escalated tickets would start from final Severity received to L1 Lead.

3.10.2 Core Functionality

i. **Escalation Mechanism**: Nodal Officers can escalate tickets by recommending them as Critical or High, based on the number of users impacted along with proper Remark.
ii. **Severity Assessment**: Any concern official from IT division (who have the rights) can review escalated tickets and determine the final severity.
iii. **Ticket Acknowledgement or further assignment**: L1 Lead would be receiving these escalated ticket and would further assign to dedicated L1 support staff if was not assigned before.
iv. **Status Monitoring**: Provides real-time status updates on all tickets.
v. **Reporting**: Generates reports on ticket status, escalation history, and resolution times.

3.10.3 Workflow



Workflow 6: Ticket Escalation by Regional Nodal Officer

i. Nodal Officers in each region view all tickets and their status.
ii. If a ticket is deemed Critical or High, the Nodal Officer escalates it by recommending the severity and noting the number of users impacted along with proper Remark.
iii. Escalated tickets are sent to the IT division for final freezing and further to L1 Lead.
iv. The L1 Lead would further assign to L1 support staff for resolution.
v. The ticket is then handled based on its final severity, with appropriate resources allocated to resolve it.
vi. Once resolved, the ticket is closed, and the status is updated in the system.

3.10.4 Role Based Access

i. **Nodal Officer:**
   a. View all tickets and their statuses
   b. Escalate tickets by marking them as Critical or High, and note the number of users impact along with proper Remark.

ii. **IT division:**
   a. View escalated tickets and their details.
   b. Review and freeze the final severity against the tickets.

iii. **L1 Lead:**
   a. View escalated tickets and their details.
   b. Further assign tickets to L1 support staff for final resolution.

4 Integration Requirements

Not Applicable

5 Out of Scope

i. Any items, features, or functionalities not explicitly covered in the Request for Proposal (RFP) would be considered out of scope for this module.

6 RFP FRS Coverage

RFP No.	RFP Description	Coverage	Covered In
K	**A dedicated Cloud based incident management tool shall be in place that offers complete solution for proactive monitoring of real-time alerts and proactive/reactive incidents.**		
K.1	System shall have a provision to raise Incidents with selection of required parameters.	Covered	Section 3.1
K.2	The system shall be able to display the real-time incident status of issues raised by users such as open tickets, assigned tickets, closed tickets along with other status on the dashboard against their incident reference number and description.	Covered	Section 2.1.xii, 3.7.2.i.c
K.3	Custom-based service reporting functionalities shall be available in the tool.	Covered	Section 2.6
K.4	System shall have dedicated two/three-level escalation matrix whereby highest escalation points shall be client itself.	Covered	Section 2.1.xii
K.5	Documented procedures, SOPs and guidelines shall be present in the repositories of the tool for reference.	Covered	Section 3.6

7 Assumptions & Dependencies

7.1 Operational Assumption & Dependencies:

i. Users should be pre-onboarded, and the relevant master data must be available in the system to ensure proper management of roles and rights.
ii. Timely Sign-off of the documents, so that development may start on time.

7.2 Technical Assumption & Dependencies:

i. The technical aspects for the integration and deployment of the Ticket Management module should align with those of the other Anna Darpan application modules.
ii. Dependencies on specific hosting providers or cloud services can impact the portal's deployment options.

8 Open Question

Not Applicable

9 Annexure

Not Applicable